

Executive Summary – Customer Churn Analysis

Overview

This project analyzes customer churn behavior to identify the major factors influencing customer retention and customer loss. The analysis includes customer demographics, service usage patterns, contract information, payment methods, and additional service adoption. Multiple visualization techniques such as count plots, percentage-based stacked charts, and comparative analysis were used to generate meaningful business insights.

Customer Churn Overview

The churn analysis shows that approximately **26.5% of customers have churned**, while around **73.5% of customers are retained**. This indicates that although the majority of customers continue using the service, a significant percentage of customers are leaving, which can impact revenue and long-term growth.

The goal of this analysis is to identify high-risk customer segments and understand the reasons behind customer attrition.

Key Insights from Analysis

1. Customer Tenure Impact on Churn

Customer tenure is one of the strongest indicators of churn behavior.

- Customers with **shorter tenure have a higher probability of churn** compared to long-term customers.
- New customers are more likely to leave because they may not yet have developed loyalty toward the service.
- Long-term customers show better retention, indicating that customer relationship building plays an important role.

Business Insight:

Improving onboarding experience, providing early engagement offers, and proactive support during the first months can reduce customer churn.

2. Internet Service Analysis

The Internet Service analysis shows different churn patterns across service types.

- Customers using **Fiber Optic internet service show higher churn compared to DSL users**.
- Fiber Optic customers contribute a significant portion of churn, indicating possible concerns related to:

- Pricing
- Service quality expectations
- Customer satisfaction

Business Insight:

The company should investigate Fiber Optic customer experience and provide better support, pricing plans, or service improvements.

3. Impact of Additional Services on Churn

The analysis of services such as:

- Online Security
- Online Backup
- Device Protection
- Tech Support

shows a clear relationship between service adoption and retention.

Customers without additional services have a higher churn percentage, while customers using these services show stronger loyalty.

Example:

- Customers without Tech Support show higher churn rates.
- Customers using protection and support services are more likely to continue their subscription.

Business Insight:

Promoting value-added services can increase customer engagement and reduce churn.

4. Contract Type Analysis

Contract duration has a strong impact on customer retention.

- Customers with **month-to-month contracts have the highest churn percentage.**
- Customers with **one-year and two-year contracts show significantly lower churn rates.**

This indicates that customers with longer commitments are more stable and less likely to leave.

Business Insight:

The company should encourage customers to move from monthly plans to longer contracts by offering:

- Discounts
- Loyalty benefits

- Contract upgrade incentives
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5. Payment Method Analysis

Payment method analysis shows different churn patterns among customers.

- Customers using certain payment methods show higher churn behavior.
- Electronic payment users appear more likely to churn compared to customers using automatic payment methods.

Business Insight:

Encouraging automatic payment methods and improving payment convenience may improve customer retention.

6. Senior Citizen Analysis

The Senior Citizen analysis shows that customer age group has an impact on churn.

- Senior customers show a comparatively higher churn percentage.
- Non-senior customers represent the larger customer base but have lower churn proportion.

Business Insight:

Personalized plans, better customer support, and simplified service options can help improve retention among senior customers.

7. Phone Service and Multiple Lines

Phone service features have a comparatively lower impact on churn.

- Customers with and without Phone Service show similar churn patterns.
- Multiple Lines customers do not show a major difference compared to single-line customers.

Business Insight:

Churn reduction strategies should focus more on service quality, contracts, and customer experience rather than phone features.

Overall Business Recommendations

Based on the analysis, the main churn drivers are:

1. **Short customer tenure**
2. **Month-to-month contracts**
3. **Lack of additional services**

4. **Fiber Optic customer dissatisfaction**

5. **Certain payment behaviors**

To reduce churn, the company should:

- Improve new customer onboarding experience.
- Provide loyalty rewards for long-term customers.
- Promote security, backup, and support services.
- Convert monthly customers into longer-term contracts.
- Identify high-risk customers early using predictive analytics.
- Improve customer support quality, especially for Fiber Optic users.

Final Conclusion

The analysis shows that customer churn is not caused by a single factor but by a combination of **customer relationship length, service adoption, contract type, and customer experience**. Customers who are more engaged with the company's services and have longer commitments demonstrate stronger retention.

By focusing on improving early customer experience, increasing service adoption, and targeting high-risk customer segments, the company can reduce churn and improve customer lifetime value.